



Chapter 15

Managing Global Systems

VIDEO CASES

Video Case 1: Daum Runs Oracle Apps on Linux

Video Case 2: Lean Manufacturing and Global ERP: Humanetics and Global Shop

Video Case 3: Monsanto Uses Cisco and Microsoft to Manage Globally



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Learning Objectives

- **What major factors are driving the internationalization of business?**
- **What are the alternative strategies for developing global businesses?**
- **What are the challenges posed by global information systems and management solutions for these challenges?**
- **What are the issues and technical alternatives to be considered when developing international information systems?**



L'Oréal's Global Makeover

- **Problem:** Large number of brands, products, locations; complex production process; multiple systems
- **Solution:** SAP's ERP system implemented globally, integrated with Apriso's FlexNet for operations management
- Demonstrates the need for global firms to have integrated systems for monitoring production
- Illustrates the use of enterprise software to enforce global quality and production standards



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Factors Driving the Internationalization of Business

- **Global economic system and global world order driven by advanced networks and information systems**
- **Growth of international trade has radically altered domestic economies around the globe**
- **For example, production of many high-end electronic products parceled out to multiple countries**
 - **For example: Apple iPhone's global supply chain**



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APPLE IPHONE'S GLOBAL SUPPLY CHAIN

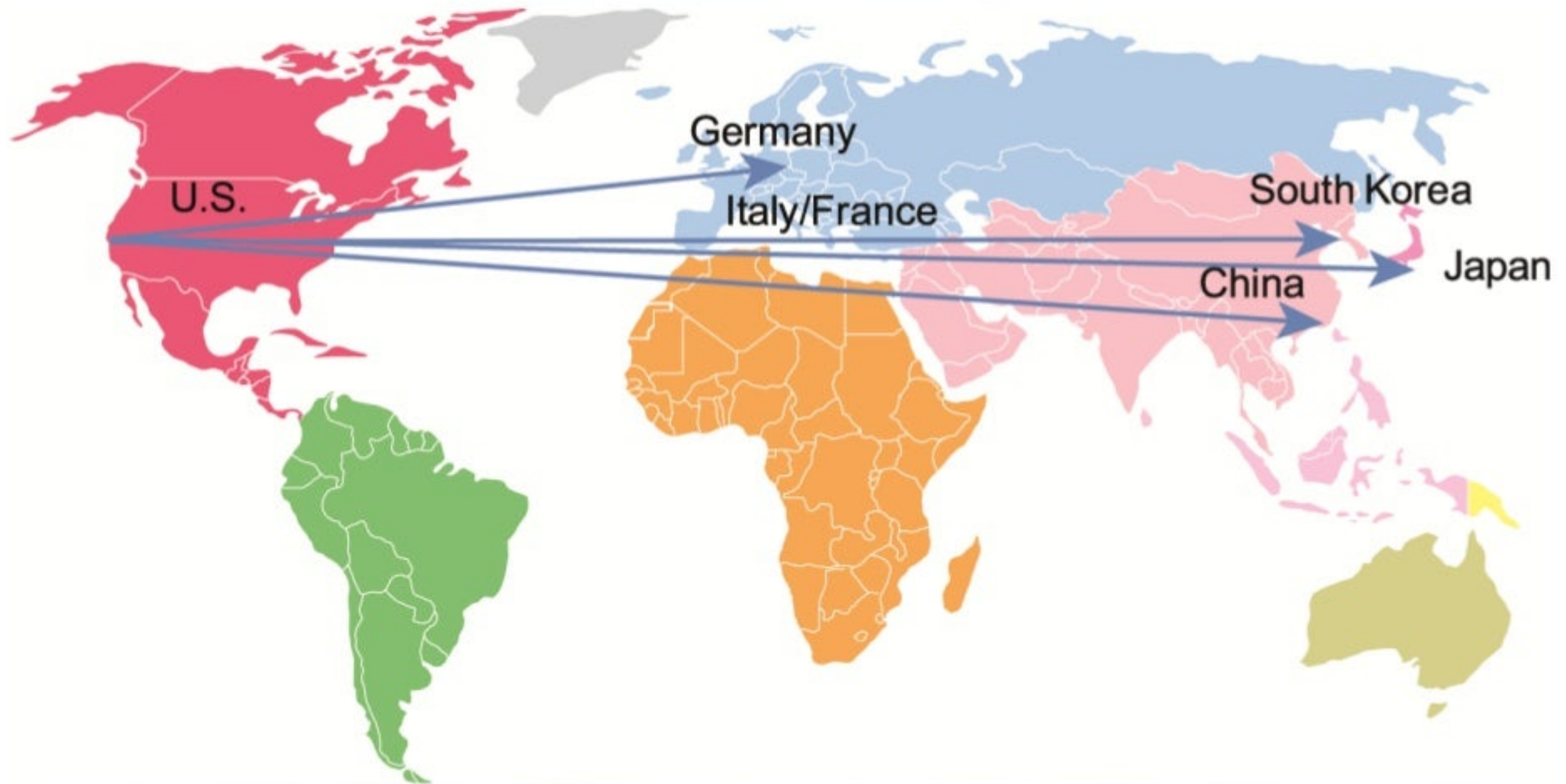


FIGURE 15-1 Apple designs the iPhone in the United States, and relies on suppliers in the United States, Germany, Italy, France, and South Korea for other parts. Final assembly occurs in China.



Factors Driving the Internationalization of Business

- **Strategy when building international systems**
 - **Understand global environment.**
 - Business drivers for global competition
 - Inhibitors creating management challenges
 - **Develop corporate strategy for global competition.**
 - **Develop organization structure and division of labor.**
 - **Consider management issues.**
 - Design of business procedures, reengineering, managing change
 - **Consider technology platform.**



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INTERNATIONAL INFORMATION SYSTEMS ARCHITECTURE

The major dimensions for developing an international information systems architecture are the global environment, the corporate global strategies, the structure of the organization, the management and business processes, and the technology platform.

FIGURE 15-2





Factors Driving the Internationalization of Business

- **Global business drivers:**
 - **General cultural factors lead toward internationalization and result in specific business globalization factors**

GENERAL CULTURAL FACTORS	SPECIFIC BUSINESS FACTORS
Global communication and transportation technologies	Global markets
Development of global culture	Global production and operations
Emergence of global social norms	Global coordination
Political stability	Global workforce
Global knowledge base	Global economies of scale



Factors Driving the Internationalization of Business

- **Challenges and obstacles to global business systems**
 - **General cultural challenges**
 - Cultural particularism
 - Regionalism, nationalism, language differences
 - Social expectations:
 - Brand-name expectations, work hours
 - Political laws
 - Transborder data flow
 - Transborder data and privacy laws, commercial regulations



Factors Driving the Internationalization of Business

- **Challenges to global business systems (cont.)**
 - **Specific challenges**
 - Standards
 - Different EDI, e-mail, telecommunication standards
 - Reliability
 - Phone networks not uniformly reliable
 - Speed
 - Different data transfer speeds, many slower than United States
 - Personnel
 - Shortages of skilled consultants



Factors Driving the Internationalization of Business

- **State of the art**
 - **Most companies have inherited a patchwork international system using traditional batch-oriented reporting, manual entry of data from one legacy system to another, and little online control and communication.**
 - **Significant difficulties in building appropriate international architectures:**
 - Planning a system appropriate to firm's global strategy
 - Structuring organization of systems and business units
 - Solving implementation issues
 - Choosing right technical platform



Strategies for Developing Global Businesses

- **Global strategies and business organization**
 - **Three main kinds of organizational structure**
 - Centralized: In the home country
 - Decentralized/dispersed: To local foreign units
 - Coordinated: All units participate as equals
 - **Four main global strategies**
 - Domestic exporter
 - Multinational
 - Franchisers
 - Transnational



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Organizing International Information Systems

GLOBAL BUSINESS AND STRATEGY

BUSINESS FUNCTION	DOMESTIC EXPORTER	MULTINATIONAL	FRANCHISER	TRANSNATIONAL
Production	Centralized	Dispersed	Coordinated	Coordinated
Finance/ Accounting	Centralized	Centralized	Centralized	Coordinated
Sales/ Marketing	Mixed	Dispersed	Coordinated	Coordinated
Human Resources	Centralized	Centralized	Coordinated	Coordinated
Strategic Management	Centralized	Centralized	Centralized	Coordinated



Strategies for Developing Global Businesses

- **Global systems to fit the strategy**
 - Configuration, management, and development of systems tend to follow global strategy chosen
 - **Four main types of systems configuration**
 1. **Centralized:** Systems development and operation occur totally at domestic home base
 2. **Duplicated:** Development occurs at home base but operations are handed over to autonomous units in foreign locations
 3. **Decentralized:** Each foreign unit designs own solutions and systems
 4. **Networked:** Development and operations occur in coordinated fashion across all units



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GLOBAL STRATEGY AND SYSTEMS CONFIGURATIONS

SYSTEM CONFIGURATION	Strategy			
	Domestic Exporter	Multinational	Franchiser	Transnational
Centralized	X			
Duplicated			X	
Decentralized	x	X	x	
Networked		x		X

FIGURE 15-3 The large Xs show the dominant patterns, and the small Xs show the emerging patterns. For instance, domestic exporters rely predominantly on centralized systems, but there is continual pressure and some development of decentralized systems in local marketing regions.



Strategies for Developing Global Businesses

- **To develop a global company and information systems support structure:**
 1. **Organize value-adding activities along lines of comparative advantage.**
 - For example: Locate functions where they can best be performed, for least cost and maximum impact.
 1. **Develop and operate systems units at each level of corporate activity—regional, national, and international.**
 2. **Establish at world headquarters:**
 - Single office responsible for development of international systems
 - Global CIO position



Challenges of Global Information Systems

- **Principle management challenges in developing global systems**
 - Agreeing on common user requirements
 - Introducing changes in business processes
 - Coordinating application development
 - Coordinating software releases
 - Encouraging local users to support global systems



Challenges of Global Information Systems

- **Typical scenario: Disorganization on a global scale**
 - Traditional multinational consumer-goods company based in United States and operating in Europe would like to expand into Asia
 - World headquarters and strategic management in United States
 - Only centrally coordinated system is financial controls and reporting
 - Separate regional, national production and marketing centers
 - Foreign divisions have separate IT systems
 - E-mail systems are incompatible
 - Each production facility uses different ERP system, different hardware and database platforms, and so on



Challenges of Global Information Systems

- **Global systems strategy**
 - **Share only core systems**
 - Core systems support functionality critical to firm
 - **Partially coordinate systems that share some key elements**
 - Do not have to be totally common across national boundaries
 - Local variation desirable
 - **Peripheral systems**
 - Need to suit local requirements only



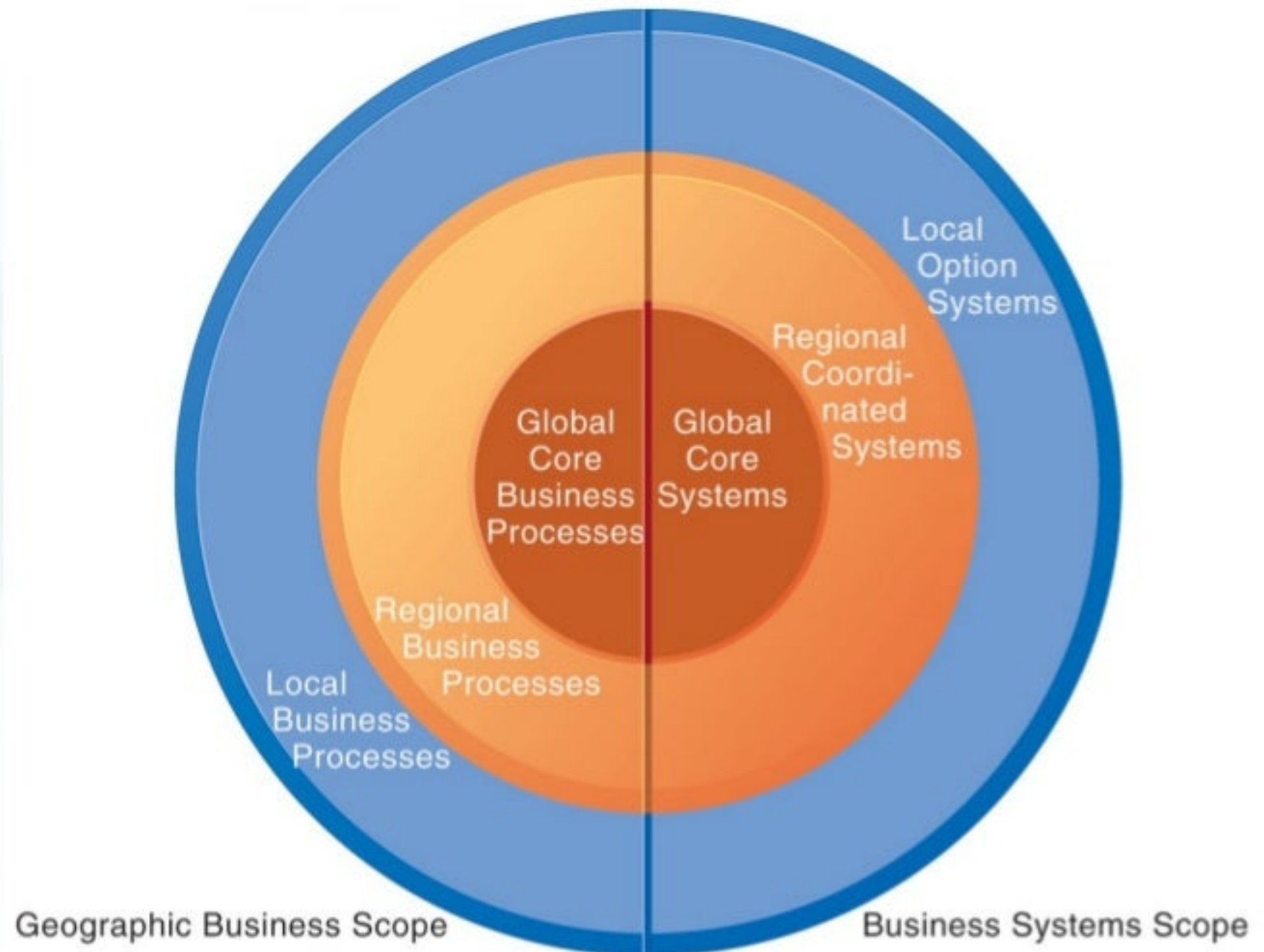
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LOCAL, REGIONAL, AND GLOBAL SYSTEMS

Agency and other coordination costs increase as the firm moves from local option systems toward regional and global systems. However, transaction costs of participating in global markets probably decrease as firms develop global systems. A sensible strategy is to reduce agency costs by developing only a few core global systems that are vital for global operations, leaving other systems in the hands of regional and local units.

FIGURE 15-4





Challenges of Global Information Systems

- 1. Define core business processes.**
 - 2. Identify core systems to coordinate centrally.**
 - 3. Choose an approach:**
 - Piecemeal and grand design approaches tend to fail.
 - Evolve transnational applications incrementally from existing applications.
-
- 1. Make benefits clear:**
 - Global flexibility
 - Gains in efficiency
 - Global markets and larger customer base unleash new economies of scale at production facilities
 - Optimizing corporate funds over much larger capital base



Challenges of Global Information Systems

- **The management solution: Implementation**
 - **Agreeing on common user requirements**
 - Short list of core business processes
 - Develop common language, understanding of common elements and unique local qualities
 - **Introducing changes in business processes**
 - Success depends on legitimacy, authority, ability to involve users in change design process
 - **Coordinating applications development**
 - Coordinate change through incremental steps
 - Reduce set of transnational systems to bare minimum



Challenges of Global Information Systems

- **The management solution (cont.)**
 - **Coordinating software releases**
 - Institute procedures to ensure all operating units update at same time
 - **Encouraging local users to support global systems**
 - Cooptation: Bringing the opposition into design and implementation process without giving up control over direction and nature of the change
 - Permit each country unit to develop one transnational application
 - Develop new transnational centers of excellence



Issues and Alternatives in Developing International Systems

- **Technology challenges of global systems**
 - **Computing platforms and systems integration**
 - How new core systems will fit in with existing suite of applications developed around globe by different divisions
 - Standardization: Data standards, interfaces, software, and so on
 - **Connectivity**
 - Internet does not guarantee any level of service
 - Many firms use private networks and VPNs
 - Low penetration of PCs, outdated infrastructures in developing countries



Interactive Session: Organizations

E-commerce Russian-Style

Read the Interactive Session and discuss the following questions

- Describe the technical and organizational obstacles to e-commerce growth in Russia.
- How do these technical and organizational factors hamper companies from doing business in Russia or setting up Russian e-commerce sites?
- Will non-Russian companies such as Amazon.com and eBay flourish in Russia? Explain.



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Interactive Session: Management

South Korea's Restricted Internet

Read the Interactive Session and discuss the following questions

- Why is the Internet restricted in South Korea?
- What are the implications of South Korea's Internet restrictions for international companies doing business in South Korea?
- What management, organization, and technology factors should be taken into account when developing an e-commerce presence for the South Korean market?



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INTERNET POPULATION IN SELECTED COUNTRIES

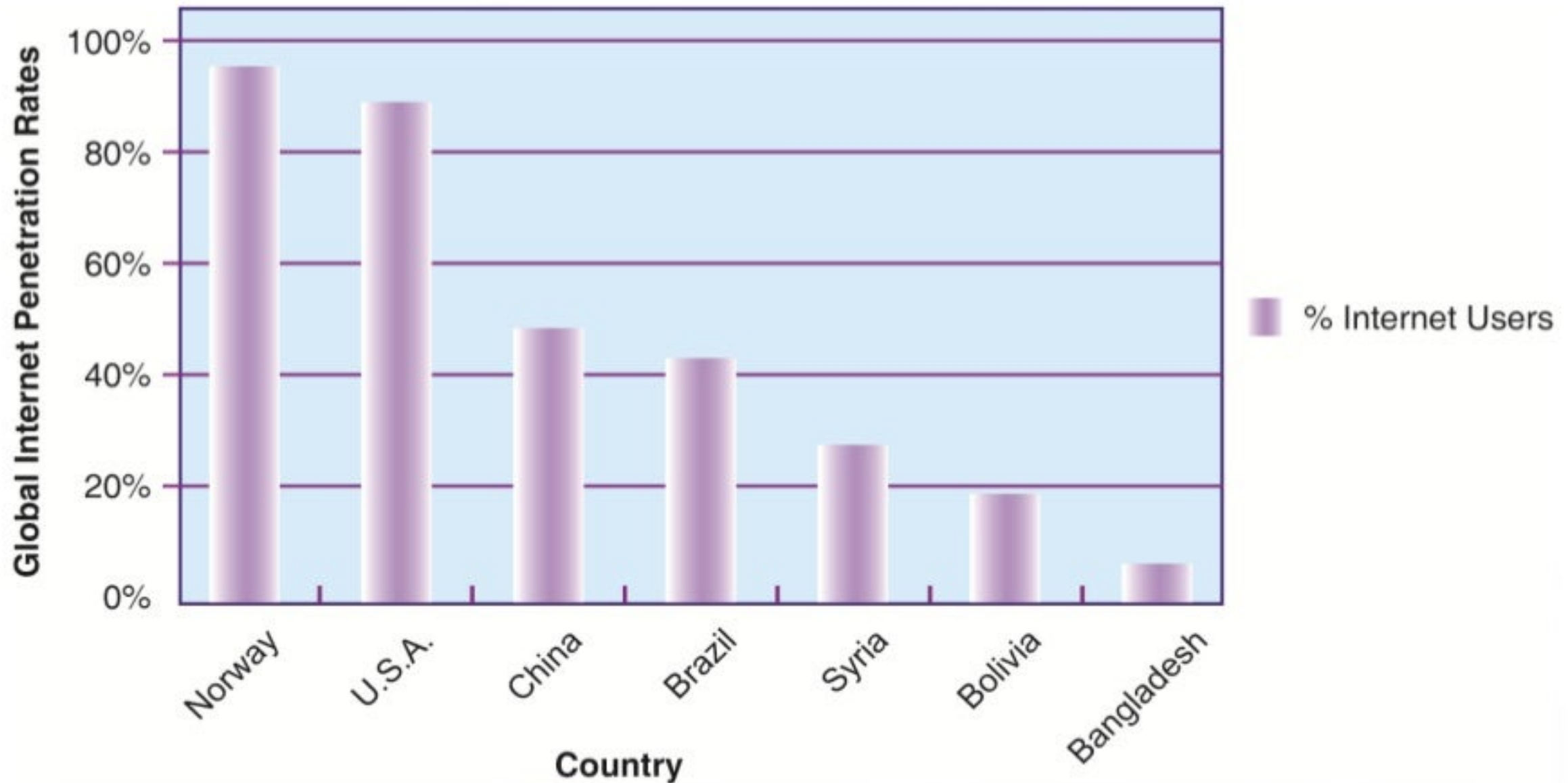


FIGURE 15-5 The percentage of the total population using the Internet in developing countries is much smaller than in the United States and Europe, but it is growing rapidly.



Issues and Alternatives in Developing International Systems

- **Technology challenges of global systems (cont.)**
 - **Software**
 - Integrating new systems with old
 - Human interface design issues, languages
 - **Software localization:** converting software to operate in second language
 - Most important software applications:
 - TPS and MIS
 - SCM, EDI, and enterprise systems
 - Collaboration tools, e-mail, videoconferencing